

Reduced coordinates for the Collatz map: exact per-step laws, anchor dynamics, and the limits of counting arguments for cycles

Ben Macindoe*

July 2026 — draft

Abstract

We study the Collatz dynamics in a reduced coordinate system that compresses each deterministic valuation run into a single block, producing a self-map F on states (ω, d) — an odd core prime to 3 and a depth. The reduction is faithful: the Collatz conjecture is equivalent to every F -orbit reaching $(1, 1)$, with nontrivial cycles in bijection. In these coordinates the per-step arithmetic is exactly solvable. A single 2-adic quantity, the *anchor* $M(\omega) = -2 \log \omega / \log 9 \in \mathbb{Z}_2$, governs the step: the exit valuation obeys the global law $s = 2 + v_2(d - M(\omega))$ on the lifting classes and is constant on the others; the depth evolution closes exactly on every residue class; and the anchor increment along one step obeys an exact law modulo any power of 2, computable from graded residues of the state. A finite window of digits consequently decides each step in an error-free trichotomy, while a digit-budget argument shows no bounded window can decide infinite horizons — localizing the full difficulty of the problem in the digit supply of the anchors. On the cycle side, a one-line elimination identity yields short rederivations of the classical exclusions for cycles of one, two, and three blocks. Our main new theorem is a sharp dichotomy for counting arguments: a trim uniform in the number of blocks p exists, giving effective finiteness at every period, but its constant necessarily degrades like $(\log_2 3)^{-p}$ — an explicit family of near-counterexamples (*staircases*: geometric climbs closed by a single crash, precisely divergent-orbit profiles bent into loops) shows size-counting cannot do better, so uniform cycle exclusion requires arithmetic (divisibility) input, not sharper counting. Finally we state the equidistribution hypothesis implicit in the classical heuristics as a precise conjecture about an exactly computable product law, prove its consequences conditionally, and report a calibration campaign whose four apparent anomalies all dissolved under controls — one of them via an exact routing lemma that the biased estimator had been reflecting.

1 Introduction

Let $T(x) = (3x + 1)/2^{v_2(3x+1)}$ denote the odd-to-odd Collatz map on positive odd integers. The Collatz conjecture asserts that every T -orbit reaches 1. This paper develops a coordinate system in which the local arithmetic of T becomes exactly solvable, and uses it for three purposes: to prove exact per-step laws for the induced dynamics; to test how far counting arguments can

*bejemite0.o@gmail.com. Developed in an extended human–AI collaboration (Claude, Anthropic); see Appendix A. All results were independently verified numerically; code and complete research records at the repository cited in Appendix A.

go toward cycle exclusion (answer: exponentially far in the number of blocks, and provably no farther); and to state precisely the statistical hypothesis that the classical heuristics leave implicit.

The idea is to stop watching individual steps. Writing $x+1 = 2^m u$ with u odd, the next $m-1$ odd steps of T are deterministic — each converts one factor of 2 in $x+1$ into a factor of 3 — and terminate in a single cascade. Compressing this *block* and splitting $u = 3^a \omega$ with $3 \nmid \omega$ yields a state (ω, d) with $d = m + a$, and the block-to-block dynamics induce a self-map F on such states (Section 2). The compression loses nothing the conjecture depends on (Theorem 2.3).

The payoff is Section 3: the entire one-step arithmetic of F is governed by the 2-adic logarithm $M(\omega) = -2 \log \omega / \log 9 \in \mathbb{Z}_2$, which we call the *anchor*. The exit valuation, the 3-absorption, the next depth, and the low-order digits of the next anchor all obey exact laws in terms of the current anchor displacement $d - M(\omega)$. A finite window of state digits then decides the next step in a trichotomy that never errs (Theorem 3.5), and an elementary accounting argument (Proposition 3.6) shows each decision *consumes* digits irreversibly, so that bounded information cannot decide unbounded horizons. In these coordinates, the difficulty of the Collatz problem is not diffuse: it is exactly the question of where the digits of the anchors come from.

Section 4 turns to cycles, where finite data can in principle beat unbounded depth. A one-line elimination identity (Proposition 4.1) converts a hypothetical p -block cycle into p divisibility conditions on $q = 2^K - 3^n$, and short arguments rederive the classical exclusions of 1-, 2-, and 3-block cycles — results due in essence to Steiner [2] and Simons–de Weger [4], with the current record $m \leq 91$ by Hercher [5]; we claim no priority, only brevity. The main new result is Theorem 4.5 together with Theorem 4.6: a trim uniform in p exists, giving effective finiteness of the candidate set at every period, but its exponential degradation in p is *intrinsic* — the staircase family satisfies every size condition with room to spare and is killed only by divisibility. Since the staircase is precisely a divergent-orbit profile closed into a loop, the cycle half and the divergence half of the problem’s residual difficulty are, in a concrete sense, the same object.

Section 5 formalizes the statistical half. The folklore “fair-coin” heuristics (going back to Terras [1]) become, in reduced coordinates, an exact statement: orbit windows should equidistribute, *while values are large*, under an explicitly computable product law π_k whose consequences (the frequency ledger, the 3-gain rate $\frac{1}{3}$, the drift) are unconditional theorems *about* π_k . We report a calibration campaign with an unusual score: four apparent violations, four dissolutions under successively sharper controls, ending with bulk uniformity unqualified at all tested depths — and with an exact lemma (deterministic routing, Lemma 5.2) explaining how the last false anomaly aimed itself at a single cell.

Related work and provenance

Block-like decompositions and stochastic models are classical (Terras [1]; Everett; see Lagarias [6] for the literature), and the 2-adic conjugacy of the Collatz map to the shift underlies all Haar-genericity heuristics. Two-adic and Diophantine methods dominate the cycle literature [2, 4, 5, 3]; effective linear forms in logarithms [7, 8, 9] are the engine, and the verified range of the conjecture currently stands at 2^{71} [10]. Contemporary with this work, a documented human–LLM exploration [11] independently develops a burst–gap decomposition and an orbit equidistribution conjecture; it contains no anchor machinery, exact global laws, or cycle apparatus, and we became aware of it only during the final literature review. The present framework originated in a hand-drawn note of 13 March 2023 — containing the odd core, the block cascade,

the base-9 structure, and the question answered by Section 3 — and was developed over three years, latterly in an extended AI collaboration documented in Appendix A.

2 The reduced system

Definition 2.1. A reduced state is a pair (ω, d) with ω odd, $3 \nmid \omega$, $d \geq 1$. Set

$$A = 3^d \omega - 1, \quad s = v_2(A), \quad x_{\text{exit}} = A/2^s, \quad C = A + 2^s, \\ \sigma = v_2(C), \quad a_+ = v_3(C), \quad m_+ = \sigma - s, \quad \omega_+ = \frac{C}{2^\sigma 3^{a_+}}, \quad d_+ = m_+ + a_+,$$

and define the reduced map $F(\omega, d) = (\omega_+, d_+)$. The projection R sends an odd x with $x + 1 = 2^m 3^a \omega$ ($3 \nmid \omega$) to $R(x) = (\omega, m + a)$.

Every odd x determines a block of the T -dynamics: with $x + 1 = 2^m u$, the odd values rise deterministically for $m - 1$ steps (each step converting one factor of 2 of $x + 1$ into a factor of 3), then exit by the cascade dividing by 2^s . The block exit is $x_{\text{exit}}(R(x))$, and the projected next state is $F(R(x))$, independently of which representative of the state the orbit entered through. Full proofs of well-definedness are given in the repository (Appendix A); the statements we need are:

Proposition 2.2 (block structure). *The BlockEntry representatives of a state (ω, d) are exactly $(u, m) = (3^a \omega, d - a)$, $0 \leq a \leq d - 1$; the block from any representative runs deterministically to the exit $x_{\text{exit}}(\omega, d)$; and the sequence of reduced states visited by any T -orbit is the F -orbit of its projection.*

Theorem 2.3 (convergence translation). *$(1, 1)$ is a fixed point of F whose R -fiber is $\{1\}$. The Collatz conjecture holds if and only if every F -orbit from a valid state reaches $(1, 1)$, and R induces a bijection between nontrivial T -cycles and F -cycles other than $\{(1, 1)\}$.*

Proof sketch. The fiber statement is Proposition 2.2 with $x = 2^d \omega - 1$ representatives; both directions of the equivalence follow by tiling T -orbits with blocks, and the cycle bijection by periodicity of the deterministic entry–exit chain. (Checked exhaustively for the fiber over $x < 3 \cdot 10^6$ and along 300 random orbits.) $\square$

3 Anchor dynamics: the exact per-step laws

For $u \equiv 1 \pmod{8}$ the equation $9^t = u^{-1}$ has a unique solution $t \in \mathbb{Z}_2$ (as 9 topologically generates $1 + 8\mathbb{Z}_2$); write $N(u)$ for it, so $N(u) = -\log u / \log 9$ in the 2-adic logarithm. N is an isomorphism $(1 + 8\mathbb{Z}_2, \times) \rightarrow (\mathbb{Z}_2, +)$, and $v_2(9^t - 1) = 3 + v_2(t)$ for $t \neq 0$ (the *isometry*).

Definition 3.1 (anchor). *For every odd ω set $M(\omega) = N(\omega^2) = -2 \log \omega / \log 9 \in \mathbb{Z}_2$, well defined since $\omega^2 \equiv 1 \pmod{8}$. $M(\omega)$ is even iff $\omega \equiv \pm 1 \pmod{8}$.*

Theorem 3.2 (global valuation law). *Call (ω, d) lifting if $\omega \equiv 1 \pmod{8}$, d even, or $\omega \equiv 3 \pmod{8}$, d odd. On the lifting classes,*

$$s = v_2(3^d \omega - 1) = 2 + v_2(d - M(\omega)),$$

while on the six non-lifting residue-parity classes $s \in \{1, 2\}$ is constant, determined by $(\omega \bmod 8, d \bmod 2)$. Moreover $a_+ = 0$ iff s is odd (so the next state gains a factor of 3 exactly when s is even), and $s \leq C(\omega) (\log d)^2$ unconditionally by p -adic Baker theory [7, 8].

Theorem 3.3 (depth closure). *On every residue class the next depth closes exactly: $d_+ = m_+ + a_+$ where $a_+ = v_3(C)$ obeys an exact 3-adic absorption law and m_+ obeys the entry-depth law: on the lifting classes, with displacement ε ($= n - N(\omega)$ resp. $n - N(3\omega)$ for $d = 2n$ resp. $2n + 1$) and shift constant $\delta_s = N(1 - 2^s)$ (of exact valuation $s - 3$),*

$$m_+ = 3 - s + v_2(\varepsilon + \delta_s),$$

and off the lifting classes $m_+ = 1 + v_2(d - M(\omega))$ (the $s = 1$ classes) or $m_+ = v_2((d - 1) - M(\omega))$ (the $s = 2$ classes). The forced carry $v_2(\delta_s) = s - 3 = v_2(\varepsilon)$ reproves $m_+ \geq 1$ structurally.

Method of proof for 3.2–3.3. All statements reduce, through the isometry, to a single *target-shift lemma*: for $c \in 1 + 8\mathbb{Z}_2$, $v_2(3^{2n}\omega - c) = 3 + v_2(n - N(\omega) + N(c))$. The valuation law is the case $c = 1$; the entry-depth law is $c = 1 - 2^s$; the odd lifting component reduces to the even one via the companion parameter 3ω ; the off-lifting cases follow by a squaring trick ($v_2(X + 1) = v_2(X^2 - 1) - v_2(X - 1)$ with $X^2 \in 1 + 8\mathbb{Z}_2$). Each law was verified exhaustively (e.g. all 62,937 states $\omega < 3000$, $d < 64$ for the entry-depth law; 8,000 random states for the unified law) with zero discrepancies. $\square$

Theorem 3.4 (low-order anchor-increment law). *Let $\Delta M = M(\omega_+) - M(\omega)$. For every $k \geq 1$, $\Delta M \bmod 2^k$ is determined by, and explicitly computable from,*

$$\omega \bmod 2^{\sigma+k+2}, \quad d \bmod 2^{\sigma+k}, \quad a_+ \bmod 2^k,$$

and the stratum data (s, σ) are themselves determined by these residues. The moduli are fixed in advance on each stratum (s, m_+) .

Theorem 3.5 (one-step trichotomy). *From the depth- k window of Theorem 3.4 alone, one can determine: the exact next depth d_+ and next residue-parity class; the exact value $s_+ \in \{1, 2\}$ if the next class is non-lifting; and, if lifting, the truncation $\varepsilon_+ \bmod 2^k$ of the next displacement — yielding s_+ exactly when nonzero, and the correct report “ $s_+ \geq k + 2$: deep cascade ahead” when zero. The window never outputs a wrong value; the undecided rate under the product law is $\approx 2^{-(k+1)}$. (Verified along real orbits: 21,296 steps, 21,000 decided with zero errors, 296 flagged with zero bound violations.)*

Proposition 3.6 (digit budget). *Deciding one step at depth k consumes the state’s 2-adic data to depth $\sigma + k + 2$ (mean $\sigma \approx 4.0$ along orbits), and no bounded window regenerates it: a window of W digits supports $O(W/(k+6))$ decided steps. Bounded-window determinism along an infinite orbit is impossible in principle.*

Proposition 3.6 is the paper’s organizing negative result: it splits the residual difficulty into a statistical question for typical orbits (Section 5) and a rigidity question for cycles (Section 4), and nothing else.

4 Cycles: rederivations, a uniform trim, and its sharpness

Fix a hypothetical p -block cycle of F with entry depths $m_t \geq 1$, exit valuations $s_t \geq 1$ ($t \in \mathbb{Z}/p\mathbb{Z}$), and set $n = \sum m_t$, $K = \sum s_t + n$, $q = 2^K - 3^n$.

Proposition 4.1 (elimination identity). $q > 0$, and for every rotation r ,

$$\omega_r 3^{a_{r-1}} q = R_r := \sum_{t=0}^{p-1} 3^{M_t} 2^{S_t} (2^{s_t} - 1),$$

with M_t, S_t the rotation's partial m - and σ -sums. Hence $q \mid R_r$ and $q \leq \min_r R_r$. For the trivial cycle, $R = 4^p - 3^p = q$ identically.

Corollary 4.2 (size balance). $2^K = 3^n \prod_t (1 + \varepsilon_t)$ with $\varepsilon_t = (2^{s_t} - 1)/(3^{d_t} \omega_t)$; if every cycle element exceeds X then $0 < K \log 2 - n \log 3 < p/X$. This is the classical Baker-side engine [3], obtained in four lines.

Lemma 4.3 (ceiling). If $K > \lceil n \log_2 3 \rceil$ then some block exit is $< p/\ln 2$. Consequently $K = \lceil n \log_2 3 \rceil$ for every nontrivial cycle with $p \leq \ln 2 \cdot X_0$, where X_0 is the verified range [10] — unconditionally for $p \leq 5$ by inspection of the odd values ≤ 7 .

Theorem 4.4 (periods 1–3; rederivations). F has no nontrivial cycles of 1, 2, or 3 blocks. For $p = 1$ the fixed-point equation $\omega 3^a (2^{s+m} - 3^m) = 2^s - 1$ admits a window argument leaving at most one candidate s per m (the window's endpoint ratio is < 2); for $p = 2, 3$, budget-trim lemmas force $0 < K \log 2 - n \log 3 < 2^{c-n/5}$ resp. $2^{3-0.115n}$, exact searches to $n = 20,000$ find no solutions (the $p=3$ search collapses $\sim 10^{12}$ raw cells to 51 size-passers, none divisible), and any effective irrationality measure for $\log_2 3$ [9] excludes the tail. These results are contained in [2, 4, 5]; the derivations, not the theorems, are the contribution.

Theorem 4.5 (uniform trim). Every nontrivial p -block cycle satisfies

$$\gamma + \log_2 p > \frac{(\log_2 3 - 1)n}{(\log_2 3)^p - 1}, \quad \gamma := K - \log_2 q.$$

Consequently (with any effective measure for $\log_2 3$) the candidate set at every period is finite and explicitly bounded: $n \leq n_0(p) = O(p(\log_2 3)^p)$.

Proof sketch. For each rotation, $q \leq R_r < p 2^{\max_t e_t}$ with the exact exponent identity $e_t - K = (\log_2 3 - 1)M_t - m_r - \sum_{j>t} s_j$; the resulting per-block conditions $m_r < \gamma' + \Phi_r$ close into a cyclic max-plus recursion on arc weights $w(A) = (\log_2 3 - 1)m(A) - s(A)$, whose total is negative since $2^K > 3^n$; from the forced reset, partial sums grow at most geometrically with ratio $\log_2 3$. $\square$

Theorem 4.6 (sharpness: the staircase). The exponential degradation in Theorem 4.5 is intrinsic to size-counting. For configurations with block depths growing geometrically at ratio $\approx \log_2 3$ and unit exit valuations, closed by a single block of unit depth and maximal exit valuation, every rotation's size condition $q \leq R_r$ holds with $\gamma = O(\log p)$: e.g. at $p = 7$, $n = 94$, the staircase $m = (4, 7, 9, 15, 23, 35, 1)$ passes all seven size tests where any polynomial-in- p trim would forbid it (84 further examples at $p = 6$ alone). All such configurations fail the divisibility conditions $q \mid R_r$. Hence no counting argument uniform in p can improve the exponential constant, and uniform cycle exclusion requires the divisibility system — equivalently, rigidity of the closed anchor walk $\sum_t \Delta M_t = 0$.

Remark 4.7. The staircase profile — geometric depth growth with shallow exits — is exactly the growth regime of the size ledger for *divergent* orbits. Size analysis cannot forbid it as a cycle for the same reason drift analysis cannot forbid divergence. The two halves of the residual difficulty meet in one configuration.

5 The equidistribution hypothesis, made exact

Let π_k denote the product law on depth- k windows: ω -residues Haar-uniform among valid ones, depths following the stationary law of the exact window chain (an 8-state class skeleton with computable entries; e.g. from class $(\omega \equiv 1 \pmod{8})$, d odd) the next depth is exactly 1, and from $(\omega \equiv 5 \pmod{8})$, d odd, $P(d_+ \text{ even}) = \frac{2}{3}$ exactly). Under π_k , unconditionally: $P(s = j) = 2^{-j}$, the 3-gain rate is $\sum_{j \text{ even}} 2^{-j} = \frac{1}{3}$, and the classical drift follows — the folklore “ledger” is a theorem *about* π_k .

Hypothesis 5.1 (AEH, bulk form). *For every k and X , along the F -orbit of almost every state (natural density), the empirical distribution of depth- k windows over visits with $x_{\text{exit}} > X$ converges to π_k .*

AEH implies the ledger with error $O(2^{-k})$ via Theorem 3.5, the exact $\frac{1}{3}$ rate, and almost-everywhere contraction; it does not exclude individual staircase tails (Remark 4.7), and by Proposition 3.6 it cannot be reached by finite-window computation. Its content is a question about digits of 2-adic logarithms, older and broader than the Collatz problem.

Calibration. An eight-round campaign (per-orbit statistics, $\sim 1,600$ – $2,600$ independent orbits per cell) produced four apparent violations, each dissolved by a sharper control: an $s=3$ excess (small-start artifact), pair correlations (same), digit biases (contamination by the *bottom regime* — the fixed drainage basin of small integers, where deviations reach $z = 41$ but reflect particular numbers, not a measure), and finally a 5σ digit bias in a single cell surviving a 2^{40} cut. The last dissolved under fixed-horizon unweighted sampling (0.2503 ± 0.0015 against null 0.25, 84,739 visits) and was explained exactly:

Lemma 5.2 (deterministic routing). *For $\omega \equiv 1 \pmod{8}$, $d = 1$: the residue $\omega \bmod 32$ determines the next class exactly — $1 \mapsto \text{itself}$ (immediate self-return), $9 \mapsto (7, 1)$, $17 \mapsto (5, 1)$, $25 \mapsto (3, 1)$, the lifting class.*

Ratio estimators over correlated visit sequences inherit this routing (residue 25 leads to deep descent, hence fewer qualifying visits, hence inflated per-orbit weight), manufacturing a bias aimed at one residue of the unique class with a self-loop channel. Bulk uniformity stands unqualified at all tested depths.

6 Discussion

In reduced coordinates the Collatz problem factors cleanly. The per-step arithmetic is solved: one 2-adic logarithm governs everything local, and finite windows decide steps without error. What remains is a single question wearing two costumes: for typical orbits, whether the anchors’ digits equidistribute (Hypothesis 5.1 — statistics); for cycles, whether closed anchor walks are rigid (Theorem 4.6 — arithmetic). The staircase shows the costumes cover the same body. We see three directions with defined success criteria: rigidity statements for closed anchor walks beyond the size level (the only route past Theorem 4.6); transfer-operator analysis of the exact window chain; and, on the statistical side, any progress on digits of 2-adic logarithms of integers, a problem this paper inherits rather than creates.

A Methodology and records

This work was produced by a human–AI collaboration over three years (2023–2026), from a hand-drawn origin note (13 March 2023) through seventy-nine drafts to a research wiki whose pages carry status headers, verification records, and refuted claims with their refutation data. Every computational claim above cites a runnable script; every “proved” result was re-verified with independently written code before being so labeled; the calibration campaign’s four dissolved anomalies, and two corrected search-filter errors on the cycle side, are documented rather than smoothed over. The complete record (wiki, code, draft history, origin note) is available from the author. Precedent for documenting LLM-assisted mathematical exploration: [11].

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